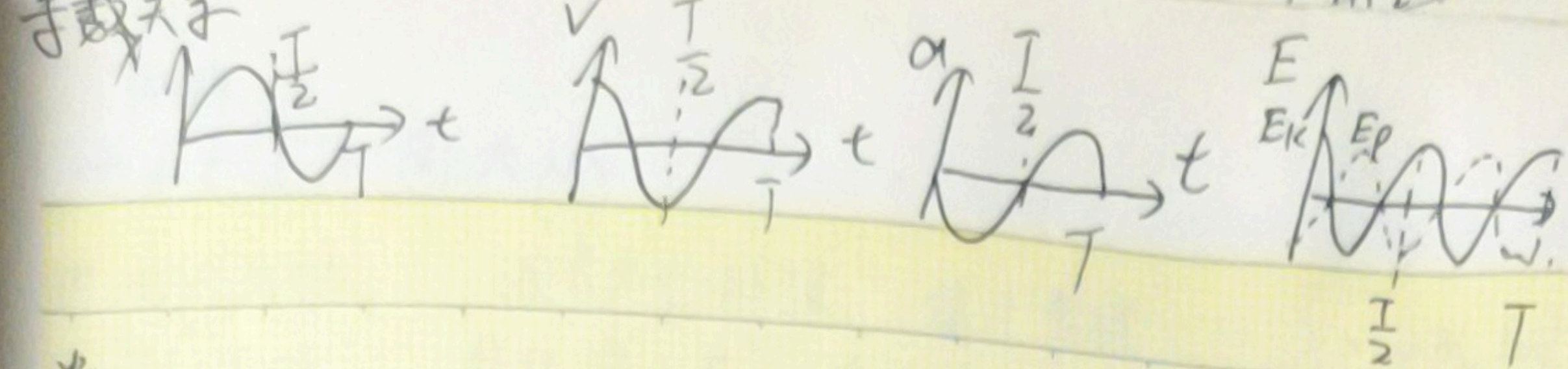
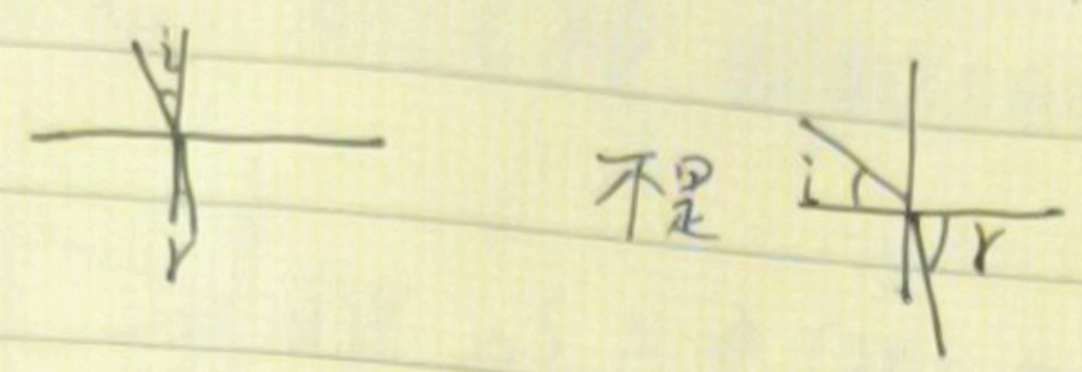




No. _____
Date. _____

光

折射定律 $n = \frac{\sin i}{\sin r}$



$$n_1 = \frac{\sin i_1}{\sin r_1}$$

$$n_2 = \frac{\sin i_2}{\sin r_2} \quad v = \frac{c}{n}$$

全反射 (光密→光疏 不折射 只反射的现象)

$\sin C = \frac{1}{n}$

$n = \frac{\sin 90^\circ}{\sin C} = \frac{1}{\sin C}$

n	$\frac{2\sqrt{3}}{3}$	$\sqrt{2}$	2	1.5	$\sqrt{3}$
C	60°	45°	30°	42°	35°

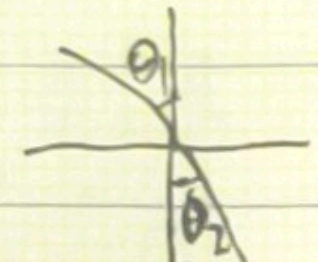
Step 1 分界面 $n = \frac{\sin i}{\sin r}$

Step 2 几何角度关联 (三角函数、外角、正弦定理、余弦定理 勾股)

反射 $\alpha = \beta$

用量角器量角度 $\cos 2\theta = 2\cos^2 \theta - 1$

$$n = \frac{n_{\text{介质1}}}{n_{\text{介质2}}} = \frac{\sin \theta_2}{\sin \theta_1} = \frac{\lambda_{\text{真空}}}{\lambda_{\text{介质水}}} = \frac{c}{v} = \frac{1}{\sin C} > 1 \quad n_{\text{介质}} > 1 \quad n_{\text{空气}} = 1$$



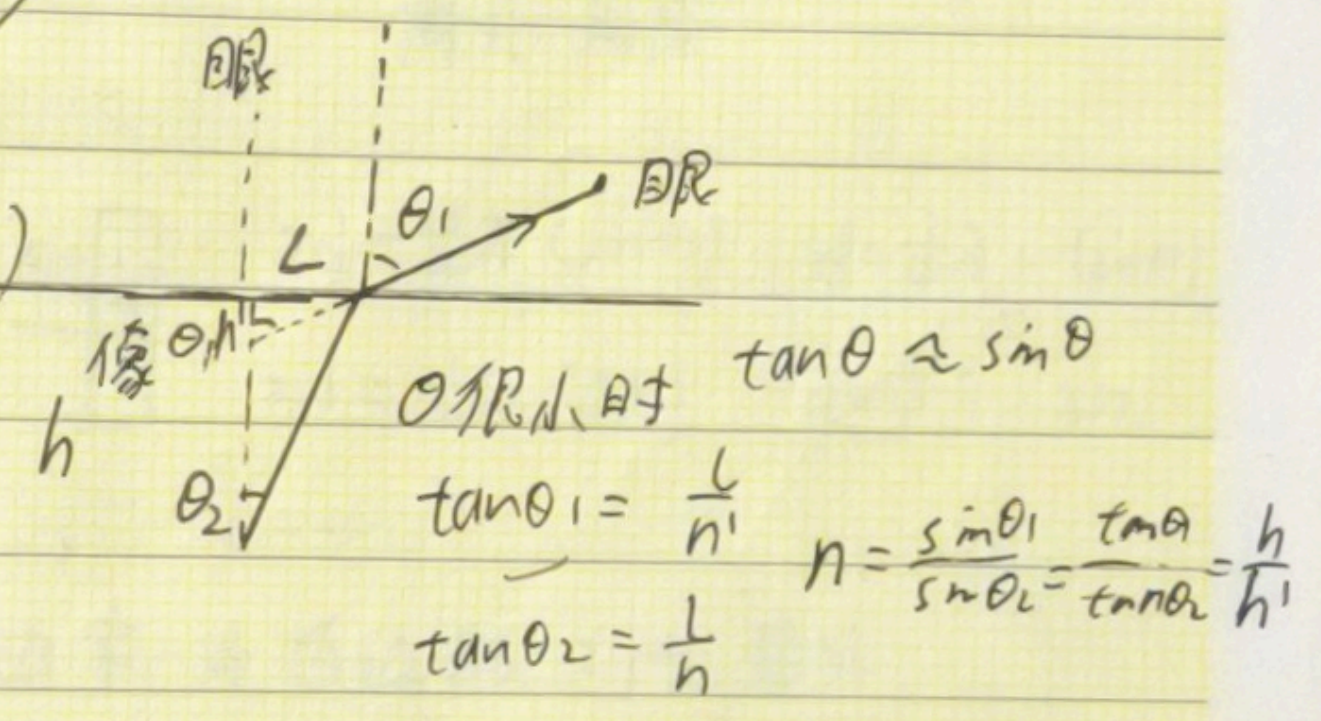
$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \text{ (完整版公式)}$$

n 与介质和光频率有关

光在 空气中传播时间 $t = \frac{s}{c}$ 介质中传播 $t = \frac{s}{v}$

水面垂直视深度 $h = \frac{h_{\text{实际}}}{n}$ (原高度除以折射率)

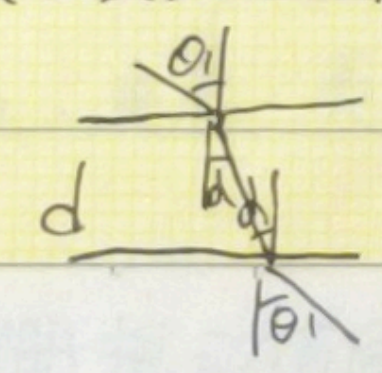
球面垂直视深度 $h' = h$ (不变)



三棱镜 两面均色散 偏折向厚边

圆镜 半径为法线 同心圆用2个半径正弦定理

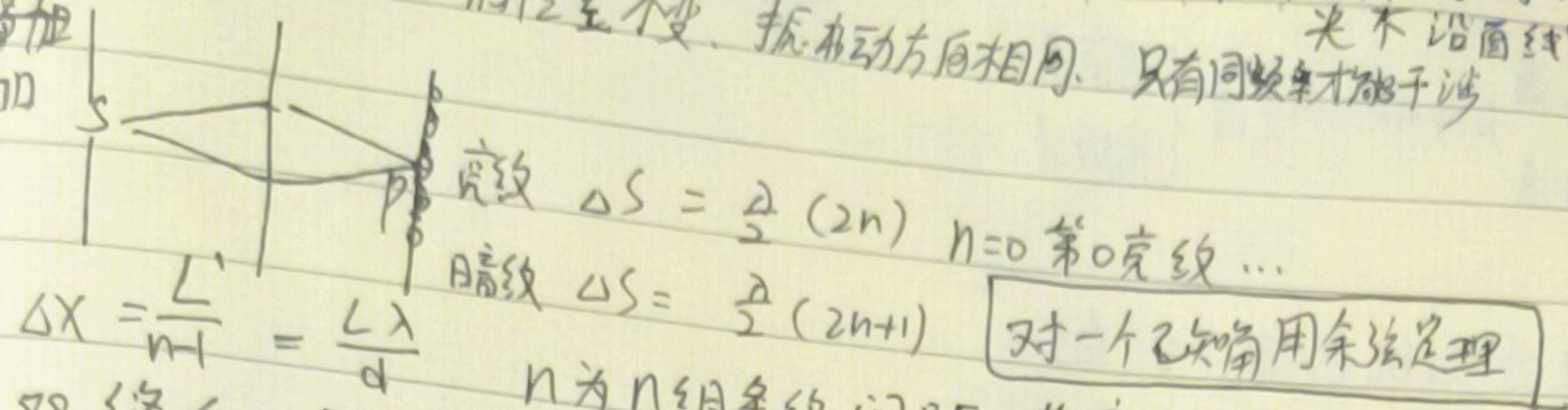
方砖玻璃 出入平行, 两面都不会全反射, 两面都色散 $S = \frac{d}{\cos \alpha} \quad t = \frac{S}{v} = \frac{dn}{\cos \alpha}$



侧移量 $x \propto \theta, d, n$ (同大)

佩尔
 $t_d + E_n$ 同大小

光的衍射：波长大于衍射强，绕过障碍， $\lambda >$ 小孔尺寸/障碍物尺寸时明显衍射
 光的干涉：同频(色)光，相位差不变，振动方向相同，只有同频率才能干涉
 同时光衍射的叠加
 同频光叠加

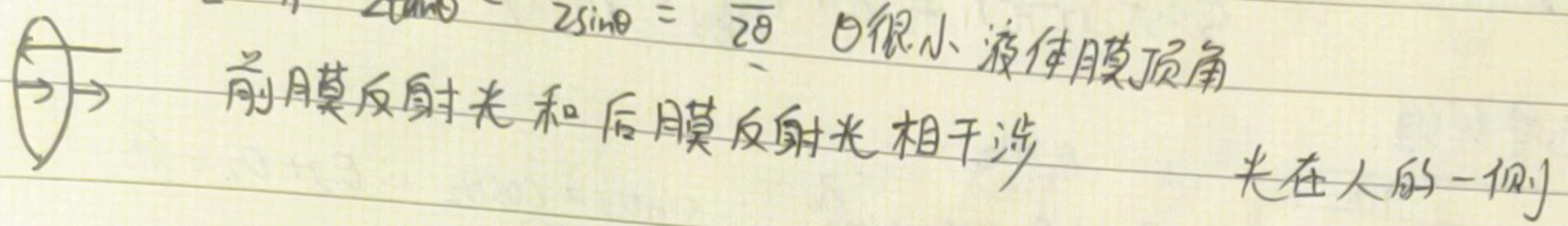


$\Delta x = \frac{L}{n-1} = \frac{L\lambda}{d}$
 n 为 n 组条纹间距 考虑单位换算数量级
 双缝向下 P 也向下 (保同时性)

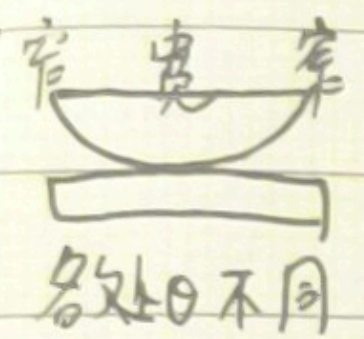
单色光 红黑等距中亮边暗
 复色光 彩色等距中白边彩 红光最外
 绿光 + 蓝光 $\Rightarrow$ 独立的蓝光条纹、绿光条纹

液体膜 (水/肥皂泡)

条纹间距 $x = \frac{\lambda}{2 \tan \theta} = \frac{\lambda}{2 \sin \theta} = \frac{\lambda}{2\theta}$ θ 很小 液体膜顶角

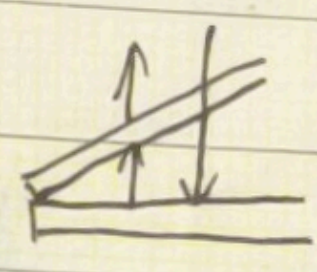


气体膜

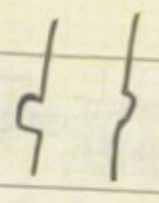


$x = \frac{\lambda}{2 \tan \theta}$

θ 大条纹变密， θ 为气体膜顶角 曲率
 中间暗，往外为亮暗相间

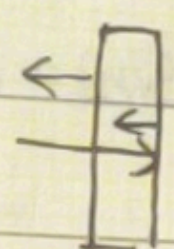


人俯视亮纹



固体膜

增透膜 (减反膜) 增强透射光，减少反射光
 增反膜 (减透膜) 增强反射光，减少透射光



$2d = \frac{1}{2}\lambda (2n+1)$ $d = \frac{1}{4}\lambda \cdot (2n+1)$
 $2d = \frac{1}{2}\lambda (2n)$ $d = \frac{1}{4}\lambda \cdot 2n$

单缝衍射、小孔衍射，中间亮两边暗，中宽边窄，中白边彩 红光最外

单缝越窄 波长越长 $\rightarrow$ 衍射越明显 $\rightarrow$ 条纹间距大 $\rightarrow$ 亮条纹范围大 $\rightarrow$ 亮条纹亮度低

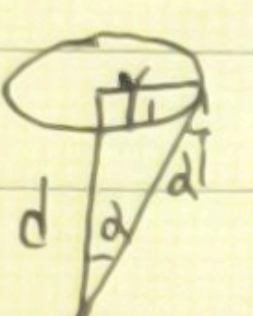
圆盘衍射 (泊松亮斑) 中窄边宽，中亮 亮暗相间

偏振：横波特有 只通过平行光栅振动方向的光
 偏振片可滤去反射光 当偏振方向与反射光的偏振方向垂直时 (横波) 中反射光
 横波 振动方向 $\perp$ 传播方向
 纵波 振动方向 $\parallel$ 传播方向

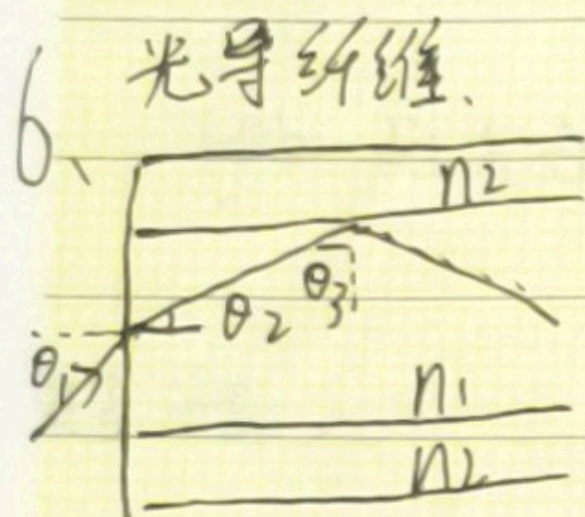
Date 无线电波 微波 红外线 可见光 紫外线 X射线 γ射线 $f \uparrow \lambda \downarrow$ 无 微 红 可 紫 X γ

连续谱 { 吸收谱 发射谱 } 反映内部结构光谱 { 连续谱 离散谱 }

1. 彩虹为光的色散 / 折射
2. 鸟看鱼偏高、鱼看鸟偏高
3. 水中唱歌 从空气到水中 声波波长 $\uparrow$, 电磁波波长短
4. 海蜃楼、车尾灯亮带 (全反射)

5.  $\sin \alpha = \frac{1}{n_1} = \frac{r_1}{\sqrt{r_1^2 + d^2}} \therefore r_1 = \frac{d}{\sqrt{n_1^2 - 1}}$ $\sin \beta = \frac{1}{n_2} = \frac{r_2}{\sqrt{r_2^2 + d^2}} \therefore r_2 = \frac{d}{\sqrt{n_2^2 - 1}}$

$$S = \pi r_2^2 - \pi r_1^2 = \pi d^2 \left(\frac{1}{n_2^2 - 1} - \frac{1}{n_1^2 - 1} \right)$$



$n_1 > n_2$

$$n_1 \sin \theta_3 = n_2 \sin \theta_2 \therefore \sin \theta_3 = \cos \theta_2 \therefore \theta_3 + \theta_2 = \frac{\pi}{2}$$

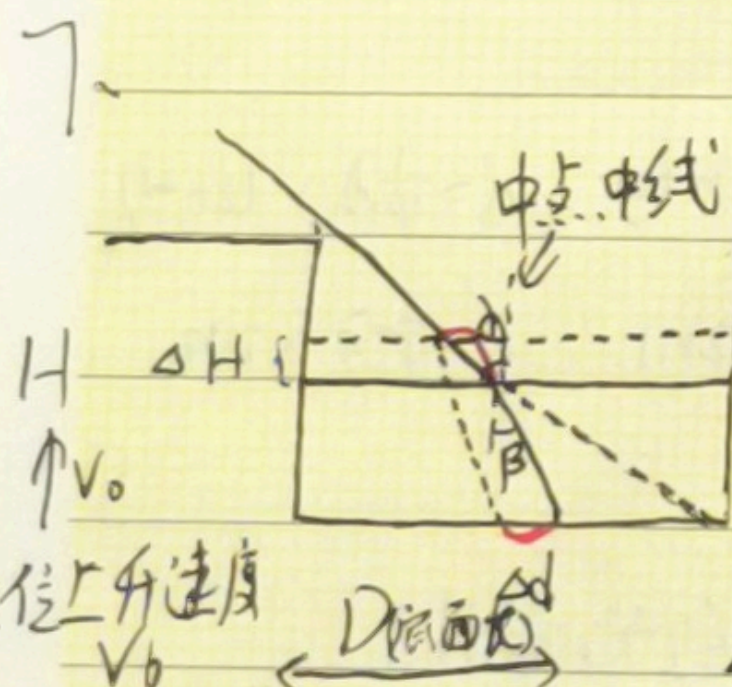
$$n_1 = \frac{\sin \theta_1}{\sin \theta_2}$$

$$\sin^2 \theta_2 + \cos^2 \theta_2 = 1 \therefore \sin \theta_1 = \sqrt{n_1^2 - n_2^2}$$

$$\frac{1}{\sin \theta_3} = n = \frac{\sin \theta_1}{\sin \theta_2}$$

$$\frac{n_1}{n_2} = \frac{1}{\sin \theta_3}$$

路程 $x = \frac{L}{\cos \theta_2}$ $V = \frac{c}{n_1}$ $t = \frac{x}{V} = \frac{n_1^2}{n_2^2} \frac{L}{c}$



① 求折射率

$$\sin \alpha = \frac{D}{\sqrt{D^2 + H^2}} \quad \sin \beta = \frac{d - \frac{D}{2}}{\sqrt{(d - \frac{D}{2})^2 + (\frac{H}{2})^2}} \quad n = \frac{\sin \alpha}{\sin \beta} = \frac{D}{2d - D} \sqrt{\frac{(2d - D)^2 + H^2}{D^2 + H^2}}$$

② 求光源与水面速度关系

$$\Delta d = \Delta H (\tan \alpha - \tan \beta) \quad ①$$

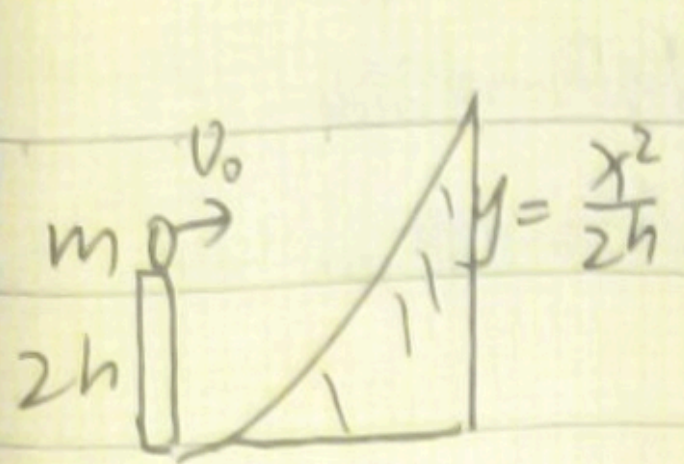
$$\tan \alpha = \frac{D}{H} \quad \tan \beta = \frac{d - \frac{D}{2}}{\frac{H}{2}} \quad \uparrow$$

$$\Delta d = \Delta H \left(\frac{2H - 2d}{H} \right) \quad ②$$

$$\frac{\Delta d}{\Delta t} = \frac{\Delta H}{\Delta t} \left(\frac{2H - 2d}{H} \right) \quad ③$$

$$\therefore V = \frac{2(d - d')}{H} V_0 \quad ④$$

交3M 用轨迹方程



求 E_{kmin}

$$y = -\frac{g}{2v_0^2} x^2 + 2h = \frac{x^2}{2h}$$

$$\therefore y = \frac{2v_0^2 h}{gh + v_0^2}$$

求表达式 $x^2 = \frac{4h v_0^2 h}{gh + v_0^2} = \frac{4v_0^2 h^2}{gh + v_0^2}$

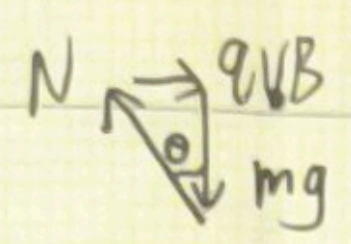
$y = \frac{1}{2h} x^2 \Rightarrow y$

$$E_k = mg(2h - y) + \frac{1}{2} m v_0^2$$

$$= \frac{2mg^2 h^2}{gh + v_0^2} + \frac{1}{2} m v_0^2$$

$$= \frac{2mg^2 h^2}{gh + v_0^2} + \frac{1}{2} m (gh + v_0^2) - \frac{1}{2} mgh$$

$$> 2\sqrt{2mg^2 h^2 \cdot \frac{1}{2} m} - \frac{1}{2} mgh = \frac{3}{2} mgh$$



球能圆周运动求 B_{min} 及对应速率

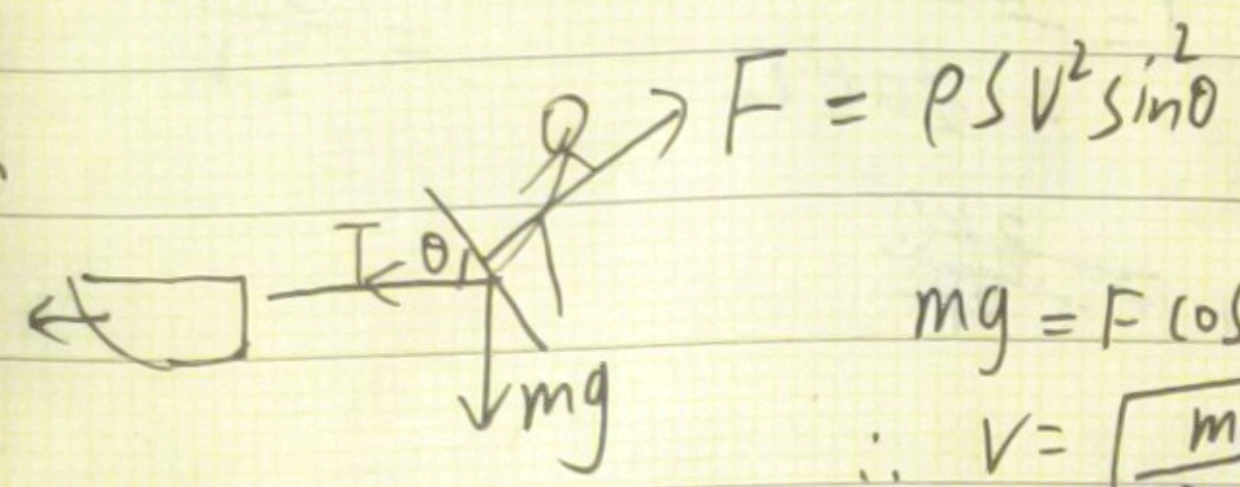
(法1) $\frac{mv^2}{r} - qVB + mg \tan \theta = 0$
 $\Delta \geq 0 \Rightarrow B$

(法2) $\begin{cases} qVB - N \sin \theta = \frac{mv^2}{r} \\ N \cos \theta = mg \end{cases}$

$$\therefore B = \frac{mv}{qV} + \frac{mg \tan \theta}{qV} \geq 2\sqrt{\frac{m^2 g \tan \theta}{q^2 V}}$$

$$\Rightarrow \frac{mv}{qV} = \frac{mg \tan \theta}{qV} \text{ 时取等}$$

3.

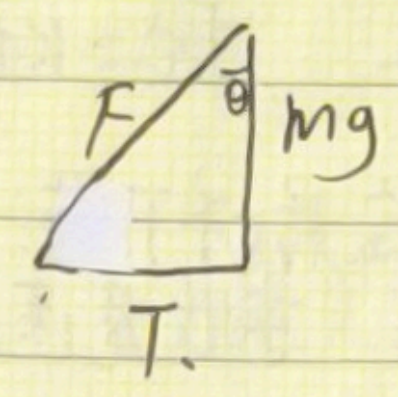


求 V_{min}

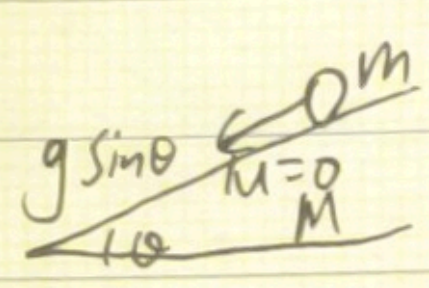
$$mg = F \cos \theta$$

$$\therefore V = \sqrt{\frac{mg}{PS \sin^2 \theta \cos \theta}}$$

$$\tan \theta = \frac{\sqrt{2}}{2} \text{ 时 } V_{min}$$



4. $\sin 2\theta = 2 \sin \theta \cos \theta$ 及 辅助角公式

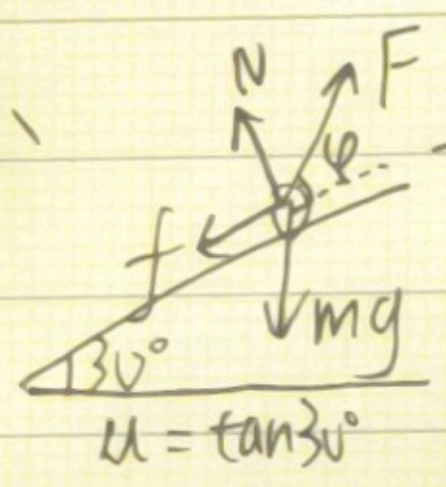


当 $\theta = ?$ 时 $f_{\pm}$ max

系统牛二律

$$f_{\pm} = M \cdot 0 + ma \cos \theta = mg \sin \theta \cos \theta \quad \theta = 45^\circ \text{ 时取极大值}$$

4.5.



$\alpha = \frac{\pi}{2}$ 求最小拉力 F

$$\begin{cases} F \cos \alpha - mg \sin 30^\circ - f = mg \\ N + F \sin \alpha = mg \cos 30^\circ \\ f = \mu N \end{cases}$$

$$F = \frac{mg \sin 30^\circ + \mu mg \cos 30^\circ + mg}{\sqrt{1 + \mu^2} \sin(\alpha + \varphi)}$$

$$\tan \varphi = \frac{1}{\mu}$$

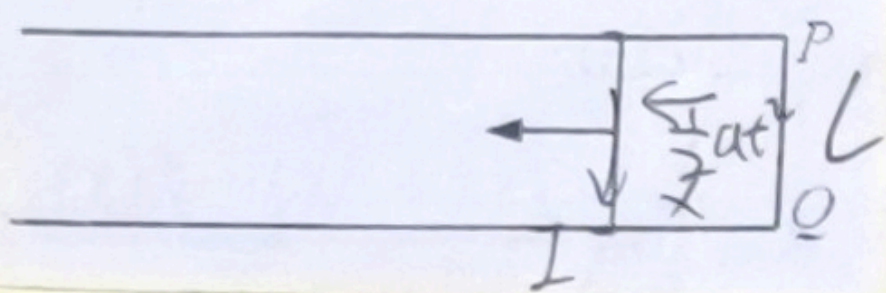
$$\Rightarrow \sin(\alpha + \varphi) = 1 \text{ 时 } F_{\text{最小}}$$

动生 + 感生 (求导) 对 t

$$\mathcal{E} = \frac{\Delta \Phi}{\Delta t} = S + BLV$$

给 $B=kt$ 直接加 $\mathcal{E} = BLV + kS$ $F_{\text{安}} = BL \frac{BLV + kS}{R}$
 没给 则按电流方向是否相同来加减 $\begin{cases} \text{同向} \rightarrow \text{相加} \\ \text{反向} \rightarrow \text{相减} \end{cases}$

B4、如图所示，两根平行金属导轨固定在水平桌面上，每根导轨每米的电阻为 $r_0 = 0.10 \Omega/\text{m}$ ，导轨的端点 P、Q 用电阻可忽略的导线相连，两导轨间的距离 $L = 0.20 \text{ m}$ 。有随时间变化的匀强磁场垂直于桌面，已知磁感应强度 B 与时间 t 的关系为 $B = kt$ 比例系数 $k = 0.020 \text{ T/s}$ 。一电阻不计的金属杆可在导轨上无摩擦地滑动，在滑动过程中保持与导轨垂直。在 $t = 0$ 时刻，金属杆紧靠在 P、Q 端，在外力作用下，杆以恒定的加速度从静止开始向导轨的另一端滑动，求在 $t = 6.0 \text{ s}$ 时金属杆所受的安培力。



$$S = Lx = \frac{1}{2}at^2 \cdot L$$

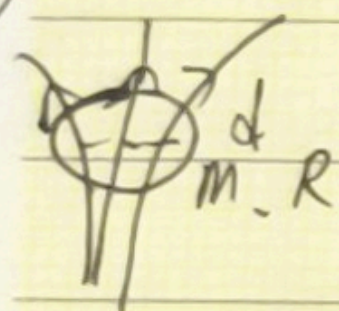
$$B = kt$$

$$v = at$$

$$I = \frac{BLat + kL \frac{1}{2}at^2}{2 \cdot \frac{1}{2}at^2 r_0} = \frac{3kL}{2r_0}$$

$$F_{\text{安}} = kL \frac{3kL}{2r_0} L = \frac{3k^2 L^2}{2r_0} t = 1.44 \times 10^{-3} \text{ N} = \frac{3kL}{2r_0}$$

(2)



金属环下落

求收尾速度

重力势能 $\rightarrow$ 电能

由功能关系

合外力为 0 X

$$B_y = B_0(1 + ky) \quad (k > 0)$$

$$mgV_m = P_{\text{电}} = \frac{\mathcal{E}^2}{R}$$

$$\mathcal{E} = \frac{d(B_0(1 + ky) \frac{\pi d^2}{4})}{dt} = B_0 \frac{\pi k d^2}{4} V_m$$

对上式对 t 求导

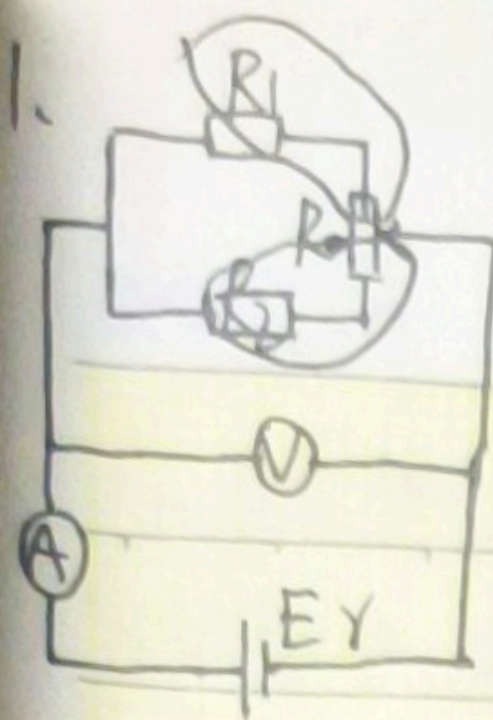
求导过程

$$\frac{d(B_0(1 + ky) \frac{\pi d^2}{4})}{dt}$$

dt

$$= \frac{B_0 k \pi d^2}{4}$$

$$\therefore V_m = \frac{16mgR}{13^2 \pi^2 k d^2}$$



$$R_{\text{总}} = \frac{(R_1 + R_2)(R_3 + R_4)}{R_1 + R_2 + R_3 + R_4} \leq \frac{(R_1 + R_2 + R_3 + R_4)^2}{4(R_1 + R_2 + R_3 + R_4)}$$

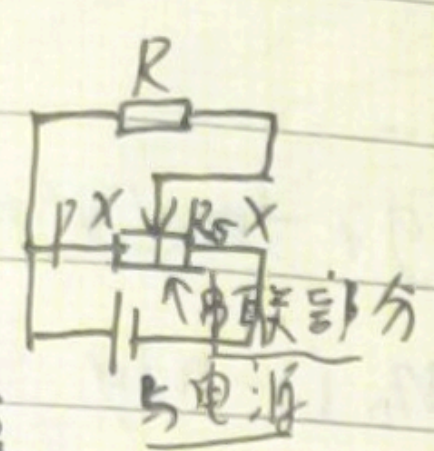
$R_{\text{等效}}$ 中间取最大

R 先 $\uparrow$ 后 $\downarrow$
 I 先 $\downarrow$ 后 $\uparrow$
 U 先 $\uparrow$ 后 $\downarrow$

当成 1 个 R 可用串反并同

当 $R_1 + R_2 = R_3 + R_4$ 能取等才先 $\uparrow$ 后 $\downarrow$

2. 传统分压



P 从左 $\rightarrow$ 右 $R_{\text{总}} \downarrow$ 与串联部分单调性一致

3. 输出功率看 P 图像



R : 滑变中值分流小 两边分流最大

$$R_{\text{外}} = \frac{RX}{R+X} + R_0 - X = R - \frac{X^2}{R+X}$$

4. 电动机

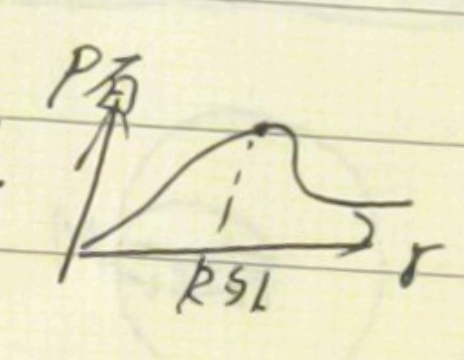
有 $R_{\text{外}}$ 与 r 关系才能定 P

效率 $\eta = \frac{U}{E} = \frac{IR_{\text{外}}}{I(R_{\text{外}} + r)} = \frac{1}{1 + \frac{r}{R_{\text{外}}}}$ 先 $\uparrow$ 后 $\downarrow$ η 与 $R_{\text{外}}$ 正相关

$P_{\text{总}} = UI$ $P_{\text{热}} = I^2 R$ $P_{\text{机}} = UI - I^2 R = Fv$ $\eta = \frac{UI}{Fv}$

5. 电源

$P_{\text{总}} = EI$ $P_{\text{有}} = UI$ $P_{\text{热}} = I^2 r$ $P_{\text{有}} = \left(\frac{E}{R+r}\right)^2 R = \frac{E^2}{R + \frac{r^2}{R} + 2r} \leq \frac{E^2}{4r}$



定向移动 10^6 m/s 热运动 10^6 m/s 电场传播 $3 \times 10^8 \text{ m/s}$

热阻 $P = P_0(1 + \alpha T)$ $R = R_1 \frac{L-x}{S} + R_2 \frac{x}{S} \dots$

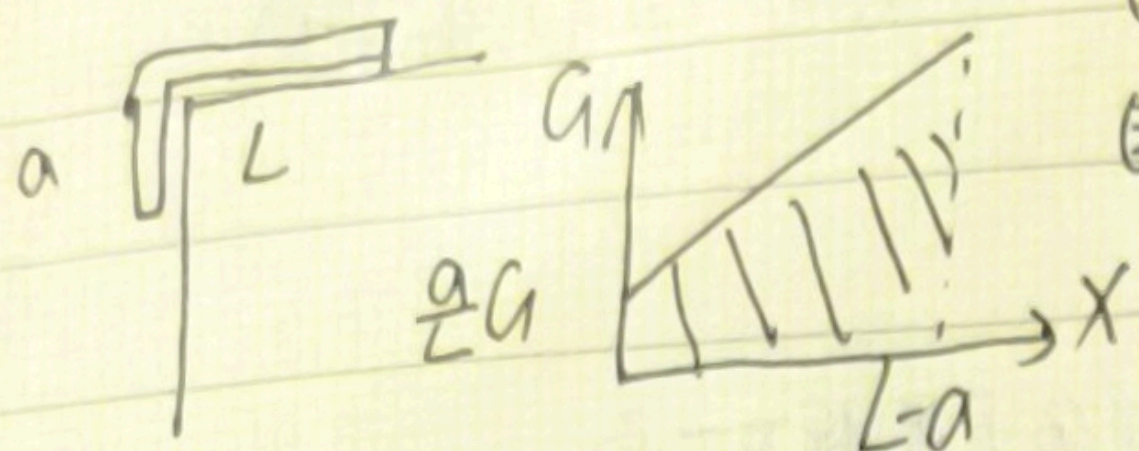
r 可以是等效电动机内阻

偏角 $(\Rightarrow)$ 电流大小

变质量功能

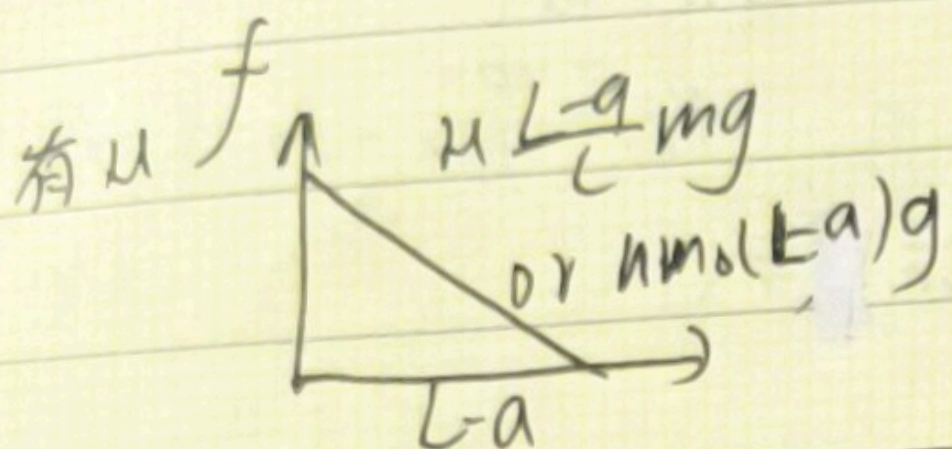
① 重心位移法 / ② 图像法 / ③ 线密度法

$$\textcircled{3} -amg\frac{a}{2} = -Lm_0\frac{L}{2}g + \frac{1}{2}m_0Lv^2$$



② 图像法

$$W = \frac{1}{2}(mg + \frac{Q}{L}mg)(L-a) = \frac{1}{2}mv^2 \therefore v = \sqrt{\frac{L^2a^2}{L}g}$$

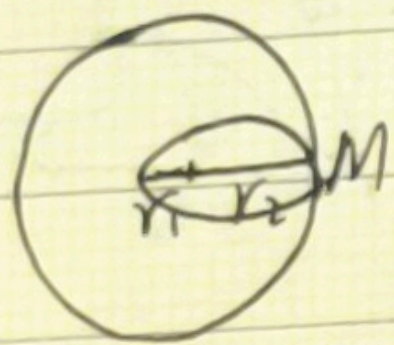


$$\textcircled{3} \text{ 加摩擦 } -am_0g\frac{a}{2} = -m_0L\cdot\frac{L}{2}g + \frac{1}{2}m_0Lv^2 + Q$$

$$W_f = \frac{1}{2}(L-a)\mu m_0(L-a)g$$

人站起来做功了 人原地走 $W_{\text{人}} = W_f$

$$\text{天体功能 } E_{p31} = -\frac{GMm}{r} \quad E_k = \frac{GMm}{2r} \quad E_{\text{机}} = -\frac{GMm}{2r}$$



$$E_{\text{机}} = -\frac{GMm}{r_1+r_2} \quad \begin{cases} \frac{1}{2}mV_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mV_2^2 - \frac{GMm}{r_2} \\ V_2 = \sqrt{\frac{2GM}{r_2(V_1+r_2)}} \quad V_2' = \sqrt{\frac{GM}{r_2}} \\ V_1 = \frac{V_2 r_2}{r_1} \\ \frac{1}{2}m\left(\frac{V_2 r_2}{r_1}\right)^2 - \frac{GMm}{r_1} = \frac{1}{2}mV_2^2 - \frac{GMm}{r_2} \end{cases} \quad W_{\text{额外}} = \frac{1}{2}mV_2'^2 - \frac{1}{2}mV_2^2$$

$$\text{冲日(追越)} \quad \left(\frac{2\pi}{T_1} - \frac{2\pi}{T_2} \right) t = 2\pi \quad \therefore t = \frac{T_1 T_2}{T_2 - T_1}$$

相向运行 ✓ 为加号

拉格朗日点 同轴转体 不会失重 地月合力 反向运行 ✓ 为减号

$$v = \omega r \quad v \propto r$$

$$a = \frac{v^2}{r} = \omega^2 r \quad a \propto r$$

$$T = \frac{2\pi}{\omega} \propto \frac{1}{\omega}$$

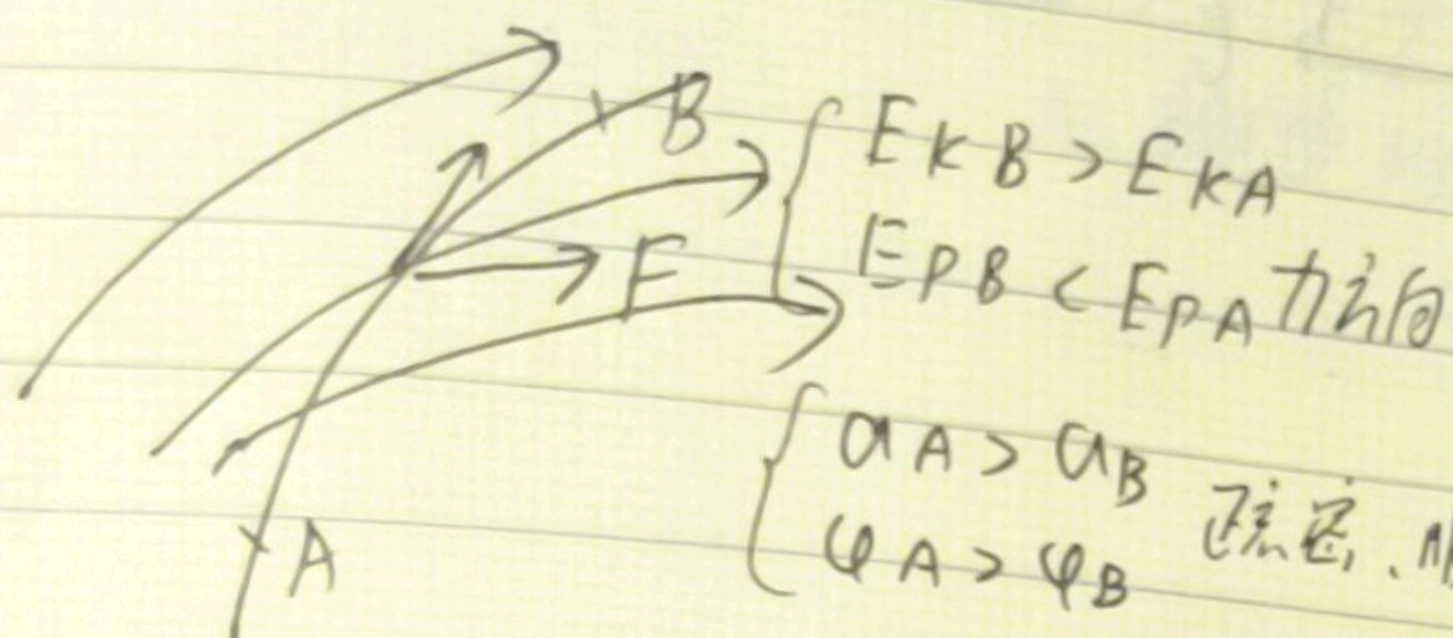
$$\omega = \frac{2\pi}{T} = 2\pi f \quad f \propto \omega$$

$$F_n = m\omega^2 r \propto m \cdot r$$

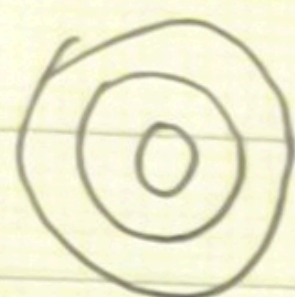
曲线运动

速度合力夹轨迹凹侧方向偏角力
 切线 V 凹侧力 锐角钝角减

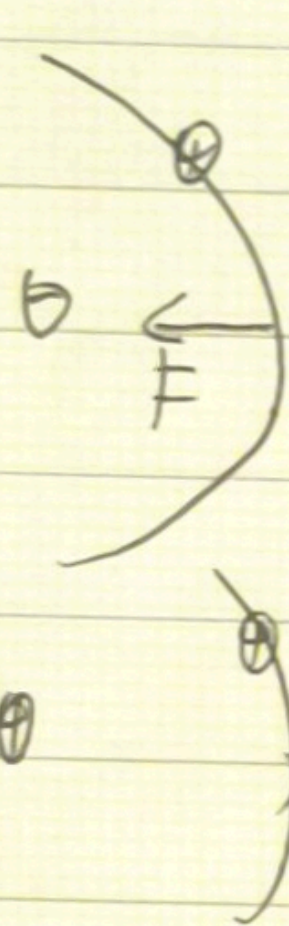
电场线切 + M
 等势面 $\perp + M$ 处处 $\perp E$, 疏密不同



电性正同反
 W 电正同反
 沿受力方向运动增势减



圆形辐状电场等势面越往外越宽, 不等间隔 (正负)

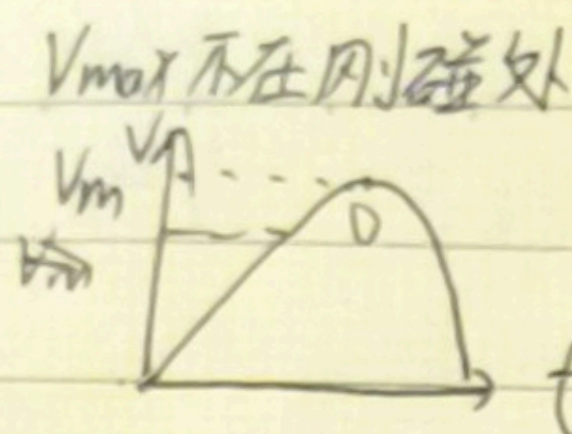


异种电荷
 凹侧包络点电荷, 离的近时动能大

粒子轨迹从X到X可以不确定

物块落在弹簧上

$0 \rightarrow Kx < mg \quad v \uparrow \quad a \downarrow$
 $0 \rightarrow Kx = mg \quad v_{max}$
 $0 \rightarrow Kx > mg \quad v \downarrow \quad a \uparrow$

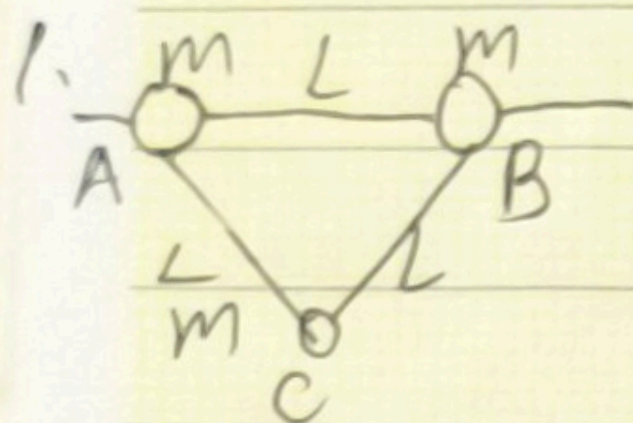


有h释放 $a > g$
 无h释放 $a = g$ 动能定理证明
 提看释放 $a < g$

变速临界 $a=0 \quad v_{max} \quad v=0 \quad a_{max}$

分离临界 $N=0 \quad a_1=a_2 \quad v_1=v_2$ 不分离 $N \geq 0$

滑动临界. 整体隔离法



① 求AB相碰瞬间速度

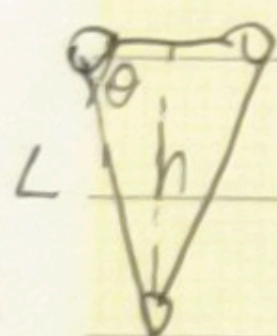
② A、B速度与C到细杆距离的关系

动能定理

$$① \quad mg(L - \frac{\sqrt{3}}{2}L) = \frac{1}{2}2mV^2$$

$$V = \sqrt{(1 - \frac{\sqrt{3}}{2})gL}$$

水平动量守恒 $mV_A + mV_B = 0$
守恒



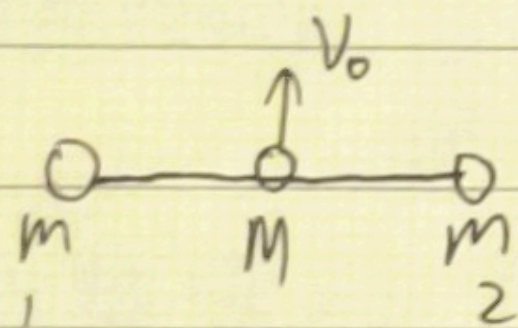
$$② \quad mg(h - \frac{\sqrt{3}}{2}L) = \frac{1}{2}2mV^2 + \frac{1}{2}mV_C^2$$

$$V \cos \theta = V_C \sin \theta$$

$$\tan \theta = \frac{h}{\sqrt{L^2 - h^2}}$$

$$\therefore V = \sqrt{\frac{gh^2(2h - \sqrt{3}L)}{L^2 + h^2}}$$

2. 求1、2相遇时速度

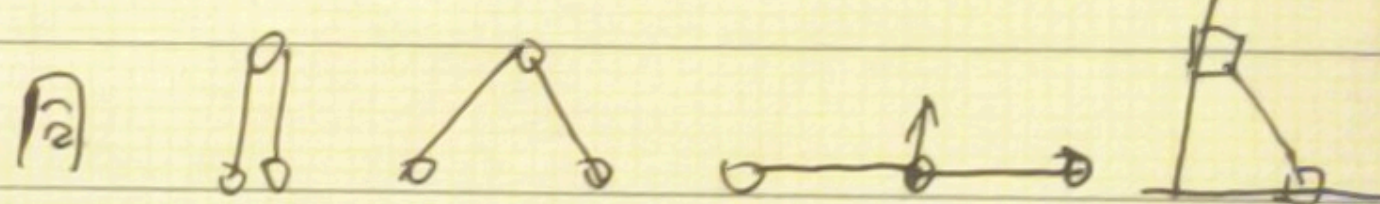
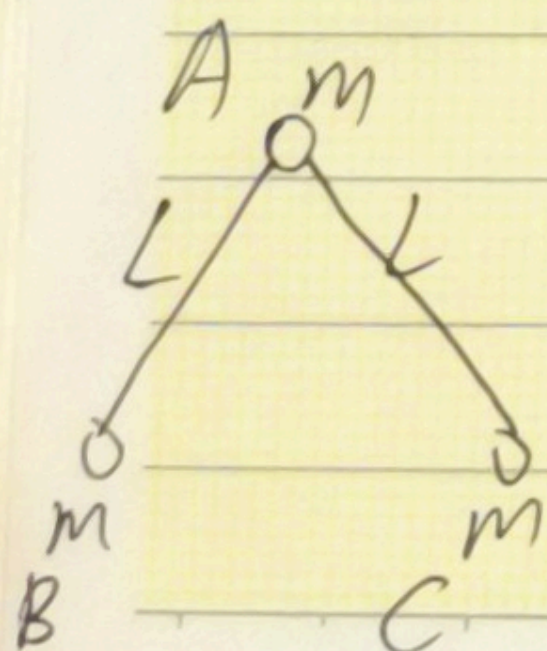


$$MV_0 = (M + 2m)V_y$$

$$\frac{1}{2}MV_0^2 = \frac{1}{2}MV_y^2 + \frac{1}{2}2m(V_x^2 + V_y^2)$$

$$V_x = \sqrt{\frac{m}{M+2m}} V_0 \quad V_y = \frac{MV_0}{M+2m}$$

$$V = \sqrt{V_x^2 + V_y^2} = \sqrt{\frac{2m(M+m)}{M+2m}} V_0$$

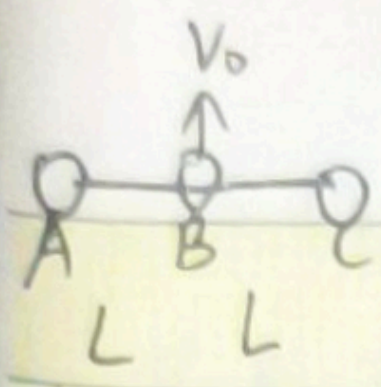


bc: $V=0$ 有 W_E $W_{负}$

F杆=0时 $a_A = g$ 其余时刻 $a_A < g$

$$V_{A \pm 0} = \sqrt{2gL} \quad mgl = \frac{1}{2}mV^2 \quad V_B = V_C = 0 \quad \text{水平动量抵消}$$

A机械能最小时 $V_A=0 \quad F_{杆}=0 \quad V_B=V_C=\frac{\sqrt{2gL}}{2}$ bc对地压力为mg
看超失重



① AC 第一次碰求 V_B

① $mV_0 = 3mV_{共}$ $V_{共} = \frac{V_0}{3}$

② 两次三球共线 V_B 、绳中拉力 F

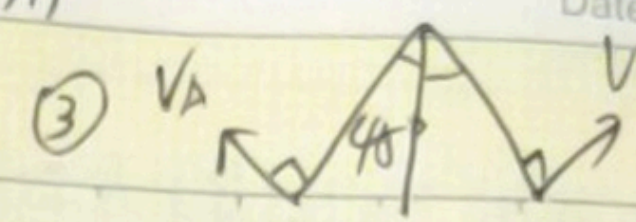
③ 运动过程中 A 最大动能和两绳夹角

②
$$\begin{cases} mV_0 = mV_B + 2mV_A \\ \frac{1}{2}mV_0^2 = \frac{1}{2}mV_B^2 + \frac{1}{2}m(V_A^2 + V_A^2) \end{cases}$$

$$\begin{cases} V_B = -\frac{1}{3}V_0 \\ V_A = \frac{2}{3}V_0 \end{cases}$$

$$F = \frac{mV_{相}^2}{L} = \frac{mV_0^2}{L}$$

AC 相对 B $V_{相} = V_0$



V_C 沿绳 V_B 方向故 A 的元分量 $V_B = 0$
 E_{KAmax}

V_A V_C 垂直绳 $\theta = 90^\circ$

$$2mAV \sin 45^\circ = mV_0$$

$$V_A = V_C = \frac{\sqrt{2}}{2}V_0$$

$\theta = 90^\circ$

$$E_{KA} = E_{KC} = \frac{mV_0^2}{4}$$